MALWARE ANALYSIS REPORT EICAR Test File Analysis

Day 06 Project 02

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1. Executive Summary

The EICAR (European Institute for Computer Antivirus Research) test file was analysed in a controlled Kali Linux sandbox environment. EICAR is not actual malware, but a standardised test string universally recognised by antivirus engines. This analysis demonstrates proper detection and analysis methodology. The sample poses no security risk and is used exclusively for testing antivirus functionality. Immediate action: none required — this is a test file only.

2. MALWARE IDENTIFICATION File Name: eicar_test.txt File Type: EICAR virus test files (detected by 'file' command) File Size: 68 bytes Cryptographic Hashes: - SHA-256: 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f - SHA-1: 3395856ce81f2b7382dee72602f798b642f14140 - MD5: 44d88612fea8a8f36de82e1278abb02f Source: Official EICAR website (https://www.eicar.org) Virus Total Detection: Detected as EICAR-STANDARD-ANTIVIRUS-TEST-FILE by all major antivirus engines (100% detection rate)

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3. STATIC ANALYSIS FINDINGS

Strings Extracted: The 'strings' utility revealed a single meaningful ASCII string:
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*

This string is the EICAR test file in its entirety. It contains no malicious code, network indicators, file paths, or registry keys. The string serves only as a standard test vector for antivirus software validation.

File Characteristics:

- No PE header (not a Windows executable)
- No embedded resources
- Plain text ASCII encoding
- No packing or encryption detected
- No suspicious entropy patterns (entropy: low, consistent with plaintext)

Analysis Conclusion:

This is not executable code. EICAR is a detection test, not a weaponised payload.

4. DYNAMIC ANALYSIS FINDINGS

Execution Attempt:

When execution was attempted (./eicar_test.txt), the shell rejected it with "permission denied" — expected behaviour for a text file. No processes were spawned, no system calls were made, and no system modifications occurred.

Process Behaviour: None observed (file is not executable)

File System Changes: None

Registry Changes: Not applicable (Linux environment)

Network Behaviour: None observed

Persistence Mechanisms: None

Defence Evasion: None

Indicators of Compromise (IOCs):

- File Hash SHA-256: 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f - String Pattern: EICAR-STANDARD-ANTIVIRUS-TEST-FILE

- File Type Signature: Matches EICAR specification

MITRE ATT&CK Mapping:

Not applicable — EICAR is not malware and exhibits no adversarial techniques.

5. DETECTION ARTEFACTS

YARA Detection Rule:

```
rule Detect_EICAR_TestFile {
  meta:
    author = "Student_2545880"
    description = "Detects EICAR antivirus test string"
    date = "2026-04-28"
    sha256 = "275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f"
  strings:
    $eicar = "X5O!P%@AP[4\\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*" ascii
    $test = "EICAR" ascii
    $antivirus = "ANTIVIRUS" ascii
  condition:
    $eicar or (all of ($test, $antivirus))
}
```

Rule Testing:

The above rule was tested against the sample using:

yara ~/yara_rules/eicar.yar ~/samples/eicar_test.txt

Result: MATCH (rule successfully detected the file)

IOC Summary Table:

Type	Value	Description
SHA-256	275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f	EICAR test file
SHA-1	3395856ce81f2b7382dee72602f798b642f14140	EICAR test file
MD5	44d88612fea8a8f36de82e1278abb02f	EICAR test file
String	X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR...	EICAR test pattern

6. CONCLUSIONS & RECOMMENDATIONS

Risk Classification: NONE — This is not malware. EICAR is a standardised test

file created specifically for validating antivirus detection capabilities.

Immediate Actions: None required.

Long-Term Defensive Improvements:

1. Ensure all antivirus/EDR (Endpoint Detection and Response) solutions are configured to detect EICAR as a baseline validation test.

2. Periodically re-test detection capabilities using EICAR to verify that security tools remain operational.

3. Establish a malware analysis sandbox (as deployed in this lab) for safe examination of suspected threats.

Limitations & Further Investigation:

This analysis covered a benign test file. Real malware analysis would include dynamic execution in isolated environments, memory analysis, network monitoring, and reverse engineering of actual code. The methodologies demonstrated here (hashing, string extraction, YARA rules, dynamic monitoring) scale directly to analysis of genuine threats.

7. REFERENCES

[1] EICAR, "EICAR Test File," [Online]. Available: https://www.eicar.org. Accessed: 2026-04-28.

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[3] V. Alvarez, "YARA: The Pattern Matching Engine," [Online]. Available:https://virustotal.github.io/yara/. Accessed: 2026-04-28.

[4] NIST, "SP 800-83 Rev. 1: Guide to Malware Incident Prevention and Handling," 2013. [Online]. Available: https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-83r1.pdf.